

Route D2-E3: E3-Instanton and Green–Schwarz Audit for ζK_{tr}

BoYu Chen

June 11, 2026

Purpose

This note tests the second Route-D string-constrained graft: whether the Route-B Majorana contact

$$\frac{M_R}{M_*} = M_V + \zeta K_{\text{tr}} \quad (1)$$

can be interpreted as an Euclidean D3-brane instanton effect in a Type IIB/F theory compactification. The goal is not to derive the benchmark value of ζ to many digits. The goal is to check whether the structure ζK_{tr} , the hidden phase/radial language, and the Majorana-only selection rule have a standard string-theory interpretation.

The answer is conditionally yes: an E3 instanton can supply a holomorphic factor

$$\zeta_\Gamma = A_\Gamma \exp(2\pi i T_\Gamma), \quad T_\Gamma = \theta_\Gamma + i t_\Gamma, \quad (2)$$

with modulus controlled by the volume-like variable t_Γ and phase controlled by the axion-like variable θ_Γ . Charged zero modes and Green–Schwarz/Stueckelberg selection rules can allow a perturbatively forbidden Majorana operator while forbidding other perturbative operators. However, the existence of the required instanton and its zero-mode spectrum is global compactification data.

1 Instanton coefficient as hidden phase/radial data

In Type IIB/F-theory language, an E3 instanton wrapping a suitable divisor or cycle Γ contributes to the superpotential with a holomorphic factor of the schematic form

$$A_\Gamma e^{-S_\Gamma + i\varphi_\Gamma}. \quad (3)$$

Equivalently, with the complexified modulus convention

$$T_\Gamma = \theta_\Gamma + i t_\Gamma, \quad (4)$$

one may write Eq. (2). The two descriptions are related by

$$S_\Gamma - \log |A_\Gamma| = 2\pi t_\Gamma, \quad \varphi_\Gamma = \arg A_\Gamma + 2\pi \theta_\Gamma. \quad (5)$$

Thus the Route-B phrase “hidden phase/radial sector” maps directly to an axion-volume pair:

$$\text{phase} \leftrightarrow \theta_\Gamma, \quad \text{radial suppression} \leftrightarrow t_\Gamma. \quad (6)$$

For the current benchmark,

$$\zeta_{\text{tar}} = 0.1076472949 + 0.0736514853 i, \quad (7)$$

so

$$|\zeta_{\text{tar}}| = 0.13043190325293763, \quad (8)$$

$$\arg \zeta_{\text{tar}} = 0.600038020318215. \quad (9)$$

If the instanton prefactor has unit modulus, the effective action needed to produce the modulus is

$$S_{\text{eff}} = -\log |\zeta_{\text{tar}}| = 2.0369040025655094. \quad (10)$$

In the convention $\zeta = \exp(2\pi iT_\Gamma)$, this corresponds to

$$\text{Im } T_\Gamma = \frac{S_{\text{eff}}}{2\pi} = 0.3241833406119675 \quad (11)$$

if $|A_\Gamma| = 1$. These are moderate nonperturbative data; they should not be confused with a small discrete clockwork prediction.

2 Majorana-only selection rule

Perturbative D-brane/F-theory vacua often contain abelian symmetries that act as selection rules on the superpotential. Some of these $U(1)$ symmetries can become massive through a Green–Schwarz/Stueckelberg mechanism, yet remain as perturbative selection rules. An instanton factor can transform under the same $U(1)$ because its axion shifts. This allows a perturbatively forbidden charged operator to appear nonperturbatively if the total charge of the instanton factor and the inserted fields is zero.

For the right-handed-neutrino Majorana operator, a minimal charge bookkeeping model is

$$q(N) = +1, \quad q(NN) = +2, \quad q(e^{-S_\Gamma + i\varphi_\Gamma}) = -2. \quad (12)$$

Then

$$q(e^{-S_\Gamma + i\varphi_\Gamma} NN) = 0, \quad (13)$$

while NN alone is perturbatively forbidden. In a microscopic instanton calculation, this charge bookkeeping is implemented by charged fermionic zero modes at the intersections of the instanton with the seven-brane stack. The zero modes must be soaked up by insertions of the desired fields. If the only available charged-zero-mode saturation pattern inserts $N_i N_j$, the instanton generates a Majorana mass but not a Dirac Yukawa or colored-triplet operator.

This is the string-theory version of the Route-B Majorana-only matching rule. It is stronger than an arbitrary EFT prohibition, but it remains conditional: the divisor Γ , its orientifold/F-theory lift, its universal zero modes, and its charged zero modes must be checked in a concrete compactification.

3 Direction in family space

Route B separates the tensor direction from the complex coefficient:

$$\Delta M_R/M_* = \zeta K_{\text{tr}}. \quad (14)$$

The E3 instanton naturally explains the coefficient as

$$\zeta = \zeta_\Gamma = A_\Gamma \exp(2\pi iT_\Gamma). \quad (15)$$

The tensor direction K_{tr} requires an additional locality or equivariance assumption. If the instanton data are blind to the detailed coordinate on the family curve and respect the same local $SL(2)$ -covariant structure used in the Route-A carrier

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \text{Sym}^2 \mathbb{C}^2, \quad (16)$$

then the only invariant symmetric tensor direction in the contact sector is the second transvectant:

$$K_{\text{tr}} \in \text{Sym}^0 \subset \text{Sym}^2(\text{Sym}^2 \mathbb{C}^2) = \text{Sym}^4 \mathbb{C}^2 \oplus \text{Sym}^0. \quad (17)$$

Thus a family-blind instanton can consistently land on K_{tr} . This is not a proof from exact global $SL(2)$ flavor symmetry. Quantum gravity does not allow exact ungauged global symmetries. The correct statement is local and spurionic: if the instanton is insensitive to the Veronese coordinate data except through the contact pairing, the unique available invariant contact direction is K_{tr} .

4 Why no ten-digit prediction follows

The instanton interpretation naturally explains the form, selection rule, and order of magnitude of ζ , but it does not compute the benchmark complex number to high precision. Multi-instanton or multi-cover corrections are expected at relative scales set by powers of the one-instanton suppression. For the benchmark,

$$|\zeta|^2 = 0.01701248138618368. \quad (18)$$

The contact-sensitivity window recorded in the Route-B benchmark scan was

$$\Delta s = 3.208798 \times 10^{-5}. \quad (19)$$

The ratio is

$$\frac{|\zeta|^2}{\Delta s} = 530.1823731560442. \quad (20)$$

Therefore, unless a concrete compactification shows that the prefactor and higher instanton corrections are exceptionally controlled, the E3 picture cannot justify a ten-digit prediction of ζ . It can justify the conditional structure

$$\zeta K_{\text{tr}} = A_{\Gamma} e^{2\pi i T_{\Gamma}} K_{\text{tr}} \quad (21)$$

and a moderate effective action $S_{\text{eff}} \simeq 2.04$.

5 Result

The D2-E3 graft is viable as a conditional string-origin statement:

$$\text{E3 instanton on } \Gamma \rightsquigarrow \zeta_{\Gamma} = A_{\Gamma} e^{2\pi i T_{\Gamma}}, \quad (22)$$

$$T_{\Gamma} = \theta_{\Gamma} + i t_{\Gamma} \rightsquigarrow \text{hidden phase/radial data}, \quad (23)$$

$$\text{GS/Stueckelberg charge shift plus charged zero modes} \rightsquigarrow \text{possible Majorana-only selection}, \quad (24)$$

$$\text{family-blind contact locality} \rightsquigarrow K_{\text{tr}} \text{ as the unique contact direction}. \quad (25)$$

It is not a completed global model. To promote this from an appendix-level interpretation to a UV completion, one must construct the instanton divisor, count universal and charged zero modes, check that unwanted operators are absent, and replay the Route-B canonical-normalization audit.

References

- [1] E. Witten, “Non-perturbative superpotentials in string theory,” arXiv:hep-th/9604030.
- [2] R. Blumenhagen, M. Cvetič, and T. Weigand, “Spacetime instanton corrections in 4D string vacua: The seesaw mechanism for D-brane models,” arXiv:hep-th/0609191.
- [3] L. E. Ibanez and A. M. Uranga, “Neutrino Majorana masses from string theory instanton effects,” arXiv:hep-th/0609213.
- [4] M. Cvetič, R. Richter, and T. Weigand, “Computation of D-brane instanton induced superpotential couplings: Majorana masses from string theory,” arXiv:hep-th/0703028.
- [5] M. Bianchi, A. Collinucci, and L. Martucci, “Magnetized E3-brane instantons in F-theory,” arXiv:1107.3732.
- [6] M. Cvetič, R. Donagi, J. Halverson, and J. Marsano, “On seven-brane dependent instanton prefactors in F-theory,” arXiv:1209.4906.